COURSE DESCRIPTION			
Program	**Architecture, Interior Design**		
Course Title	**Fundamentals of Architecture**		
Course Code	**1182**	**Semester**	**I**
Type of Course	**Practicum**	**Contact Hours**	**90**
Teaching Scheme (L: T:P:R)	**3:1:2:0**	**Credits**	**5**
CIE Marks	**40**	**ESE Marks**	**100**

Introduction

The course aims to develop an understanding of the fundamental elements and principles of visual design and architecture. It focuses on the basic techniques of drawing, painting and drafting which form the foundation of architectural representation. The subject also examines the concepts and architectural features of buildings designed by renowned Indian and international architects, providing students with a broader perspective on architectural thought and practice.

Course Objectives

1	To enable the students to identify the basic elements and principles of architecture design.
2	To introduce the fundamental techniques of visual design through drawing, painting and drafting.
3	To comprehend the basics of free hand sketching appropriate to visual and architectural representation.
4	To comprehend the concepts and architectural features of buildings designed by well known Indian and International architects.

Course Pre-requisite

Topic	Course code	Course Title	Semester
NIL			

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Apply fundamental architectural drawing skills and design principles to represent elements of design concepts through clear, accurate, and well-composed drawings.	Apply
CO2	Apply colour theory, colour schemes, colour psychology, and texture concepts to create effective visual compositions using appropriate materials and design principles.	Apply
CO3	Apply basic design principles—balance, proportion and scale, rhythm, hierarchy, contrast, harmony, and emphasis—by developing design compositions and outdoor sketches that effectively represent form, depth, lighting, and spatial quality.	Apply
CO4	Apply the design concepts and architectural features of selected Indian and international buildings and represent their form, proportions, and spatial composition through freehand sketches	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	2	-	2	-	-	-	2	-	1
CO2	2	-	3	-	2	2	-	-	2	-	1
CO3	2	1	3	-	2	2	-	1	2	-	1
CO4	3	2	3	1	2	1	-	-	2	-	2
Total	10	3	11	1	8	5	0	1	8	0	5
Correlation Level	2.5	2	2.75	1	2	1.67	-	1	2	-	1.25
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"											

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	R				CIE	ESE	Total	CIE	ESE	Total	
3	1	2	0	6.5	90	5	20	60	80	20	40	60	140

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus – Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	P
Module I	Elements of Design	Basics of architectural drawing	3	0
Module I	Elements of Design	Elements of design	6	0
Module I	Elements of Design	Application of elements	3(T)	0
Module I	Elements of Design	Representation of elements of design	0	8
Module II	Color and texture	Colour theory	6	0
Module II	Color and texture	Texture	6	0
Module II	Color and texture	Colour scheme	4(T)	0
Module II	Color and texture	Colour compositions	0	2
Module II	Color and texture	Representation of texture	0	2
Module III	Principles of Design	Principles of design	9	0
Module III	Principles of Design	Illustration of principles of design	3(T)	0
Module III	Principles of Design	Representation of principles of design	0	6
Module III	Principles of Design	Freehand sketching	0	4
Module IV	Famous architectural buildings	Introduction of famous Indian & International buildings	12	0
Module IV	Famous architectural buildings	Design features and significance	4(T)	0
Module IV	Famous architectural buildings	Free hand sketching of buildings	0	12

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/P)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Basics of architectural drawing	Explain the sheet layout, line weight, scale and lettering	L	CO1	Understand	3
Elements of design	Describe the Primary elements of design - Point, Line, Plane, Volume - Characteristics and uses, Form- Definition, visual and relational properties, regular and irregular forms.	L	CO1	Understand	6
Application of elements	Analyze the primary elements of design by explaining their characteristics, functions, and applications in architectural design.	T	CO1	Understand	3
Representation of elements of design	Demonstrate different types of Lines, study its characteristics and Visual Impacts (Vertical, Horizontal, Diagonal, Zig-Zag, Curvilinear, Spiral etc.,)	P	CO1	Apply	4
Representation of elements of design	Illustrate different compositions of solids and voids and demonstrate the application of lighting and shading techniques.	P	CO1	Apply	4
Module II					
Colour theory	Describe Colour - Qualities of colour - Hue, Value& Intensity Primary, Secondary & Tertiary colours. Complimentary, Analogues, Monochromatic, Triadic, Tint & shade, Warm & cool. Psychology of colors.	L	CO2	Understand	6
Texture	Explain the different types of textures and discuss their characteristics, uses, and significance in architectural design.	L	CO2	Understand	6

Colour scheme	Understand the concept of the colour wheel, its components, and the relationships between colours.	T	CO2	Understand	4
Colour compositions	Illustrate various compositions using colour schemes (Complimentary, Analogues, Monochromatic, Triadic)	P	CO2	Apply	2
Representation of texture	Prepare compositions of textures using natural and artificial materials.	P	CO2	Apply	2
Module III					
Principles of design	Explain the basic principles of design such as Balance, Proportion & Scale, Rhythm, Hierarchy, Contrast, Harmony and Emphasis.	L	CO3	Understand	9
Illustration of Principles of design	Illustrate the basic principles of design—Balance, Proportion and Scale, Rhythm, Hierarchy, Contrast, Harmony, and Emphasis—by identifying their application in selected architectural Buildings	T	CO3	Understand	3
Representation of principles of design	Develop comprehensive design compositions demonstrating the application of Balance, Proportion & Scale, Rhythm, Hierarchy, Contrast, Harmony and Emphasis through drawings.	P	CO3	Apply	6
Free hand sketching	Sketch any nearby outdoor space and apply appropriate lighting and shading concepts to represent depth, form, and spatial quality.	P	CO3	Apply	4
Module IV					
Introduction of famous Indian buildings	Describe the design concepts and architectural features of following famous Indian buildings:- 1. Kovalam Beach Resort, Charles Correa 2. Sangath,Ahmadabad,B.V Doshi 3. Rashtrapathi Bhavan, New Delhi, Edwin Lutyens	L	CO4	Understand	6
Design features and significance	Discuss the architect's design philosophy, spatial organization, material selection, climatic response, and the architectural significance of each project.	T	CO4	Understand	2
Free hand sketching of buildings	Practice freehand sketching of the above-mentioned buildings to study and represent its architectural form, proportions, spatial composition, and distinctive features and an understanding of design concepts through manual drawing.	P	CO4	Apply	6
Introduction of famous international buildings	Describe the design concepts and architectural features of following famous International buildings:- 1. Falling Water, Pennsylvania, USA, FL Wright 2. Villa Savoye, France, Le Corbusier 3. Farnsworth House, Mies Van der Rohe	L	CO4	Understand	6
Design features and significance	Discuss the architect's design philosophy, spatial organization, material selection, climatic response, and the architectural significance of each project.	T	CO4	Understand	2
Free hand sketching of buildings	Practice freehand sketching of the above-mentioned buildings to study and represent its architectural form, proportions, spatial composition, and distinctive features and an understanding of design concepts through manual drawing.	P	CO4	Apply	6

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Maintain a sketchbook or journal documenting observations, experiments, and reflections on lines, forms, textures, and colours.	6.5
2	Practice various lettering styles and prepare your own lettering style.	6.5
3	Prepare a logo/poster incorporating the basic elements of design.	6.5
4	Create freehand sketches or small models to represent regular and irregular forms using simple materials.	6.5
Module II		
5	Collect examples of different colour schemes from magazines, newspapers, or online sources and classify them	6.5
6	Observe and analyze the colour scheme of a residential or commercial interior/exterior using photographs or site visits and prepare a report.	6.5

7	Collect photographs or samples of natural and artificial textures found in the surroundings and classify them.	6.5
8	Study selected furniture, interior spaces, or architectural elements and analyze how function, strength, and aesthetics are integrated in the design.	6.5
Module III		
9	Develop an abstract design composition demonstrating any three principles of design using basic geometric shapes.	6.5
10	Prepare a presentation by observing nearby built environments and documenting examples of interior and exterior spaces based on the principles of design.	6.5
11	Develop small physical or digital models to demonstrate proportion, scale, and balance using simple materials such as cardboard, thermocol, or digital tools.	6.5
12	Compare two or more spatial organizations and discuss their suitability for different building types and interior functions.	6.5
Module IV		
13	Compare one Indian and one international building in terms of philosophy, spatial organization, form, material use, and contextual response.	6.5
14	Prepare concept diagrams illustrating site response, circulation, massing, and relationship with nature for the selected buildings.	6.5
15	Create simple physical or digital models to understand the spatial relationships of any one selected building.	6.5
Total Self-Learning hours		97.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Graphic Thinking for Architects and Designers	John Wiley & Sons	New York, 2001
2	Design fundamentals in Architecture	V.S.Pramar	Somaiya Publications Pvt.Ltd., New Delhi
3	Principles of two - Dimensional Design	Francis D.K.Ching	Van Nostrand Reinhold Co.,(Canada)

REFERENCE			
Sl. No.	Title	Author	Publication
1	The Fundamentals of Architecture	Lorraine Farrelly	AVA Publishing, 2011

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://urb.bme.hu/segedlet/angol/intro/2009/09_intro2_construction.pdf	I & II
2	https://www.archdaily.com/	IV

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Drawing boards		30
2	Drawing sheets		30
3	Drawing table		30
4	Drawing Chair		30
5	Drawing instruments - T scale, Set square		30

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology – Practicum

Mode	Attendance	CA1	CE	CA2	CA3	CA4	CA5	ESE	
		Average of Self learning activities from each module	Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)	Lab Examination (internal)
Duration				3 hours	3 hours	2 hours	2 hours	2.5 hours	3 hours
Exam mark		50	50	40	40	50	50	60	40
Converted to		5		10	10	10	10		
Marks	5	5	10	10		10		60	40
Total	40							100	
Tentative schedule	End of semester	End of semester	End of semester	By 8th week	By 15th week	8th week	15th week	End of semester	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Regular attendance, timely submission of drawing sheets and models, preparedness for studio work, and understanding of objectives and related theory.	10
2	Design Process & Methodology	Ability to understand the given problem, apply design principles, follow a systematic approach in developing compositions, spatial diagrams, sketches, and drawings	10
3	Representation & Documentation	Accuracy, neatness, clarity of drawings, proper use of scale, line weight, lettering, color application, textures, and overall presentation quality.	15
4	Analysis & Conceptual Understanding	Ability to analyze design concepts, spatial organization, principles of design, and architectural features, supported through diagrams, sketches, and brief explanations.	10
5	Workmanship, Discipline & Studio Conduct	Quality of workmanship, drawing discipline, use and care of tools, cleanliness of workspace, teamwork (where applicable), and professional attitude.	5
Total			50

Scheme of Evaluation - Practical Test

Category	Things to evaluate	Marks
Write-up / Concept Note (Before Studio Work)	Clear statement of aim and objectives, understanding of design concepts, supporting reference sketches/images, and a systematic approach to drawings or compositions.	10
Design Development & Execution	Correct application of sheet layout, line weight, scale, lettering, elements and principles of design, colour schemes, spatial organization, and freehand sketching techniques; effective handling of drawing tools.	10
Output & Presentation	Neatness, accuracy, and clarity of drawings; quality of compositions, spatial diagrams, colour wheels, textures, and sketches; correctness of representation as per expected learning outcomes.	8
Viva Voce	Conceptual understanding of interior and exterior spaces, design principles, colour theory, spatial organizations, and architectural features of famous buildings; ability to justify design decisions.	8
Record / Portfolio	Completion, organization, and neatness of practical record/portfolio; proper sequencing of works from Modules I–IV and certification by faculty.	4
Total		40

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	

2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60